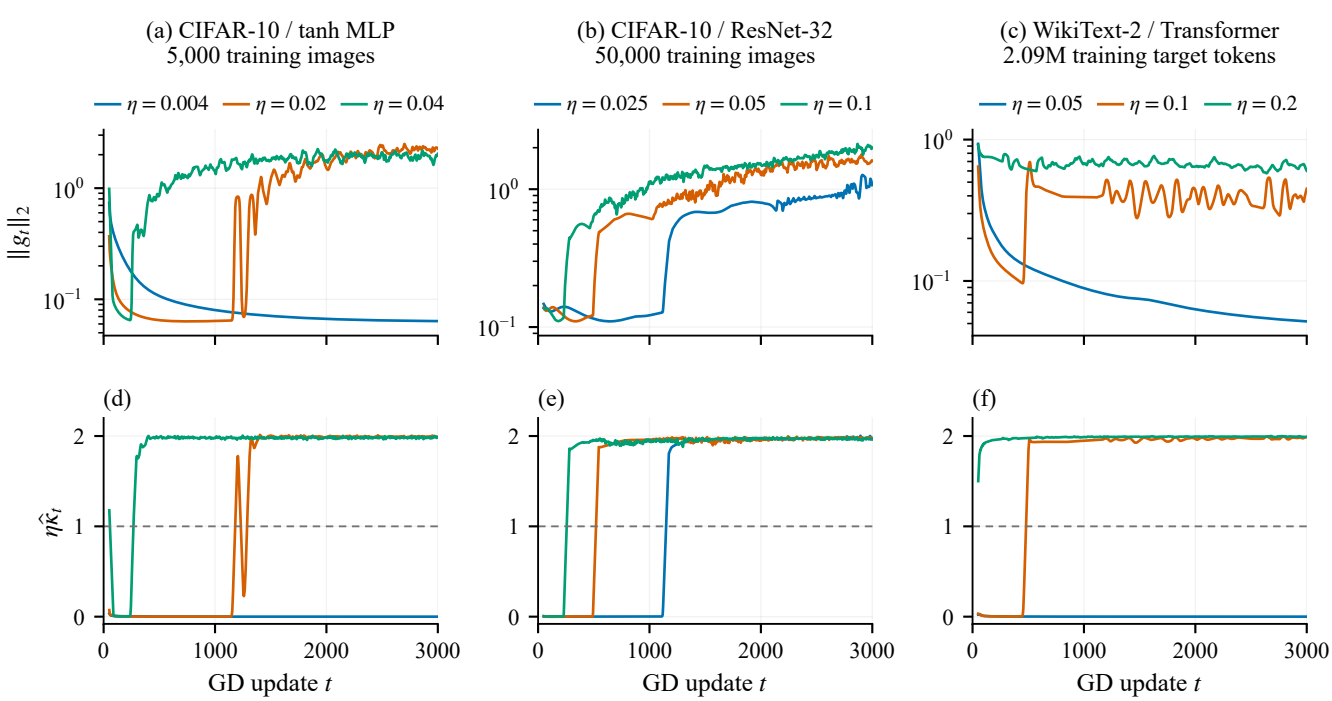

(a) CIFAR-10 / tanh MLP
5,000 training images
(b) CIFAR-10 / ResNet-32
50,000 training images
(c) WikiText-2 / Transformer
2.09M training target tokens
η = 0.004
η = 0.02
η = 0.04
η = 0.025
η = 0.05
η = 0.1
η = 0.05
η = 0.1
η = 0.2
‖g_t‖_2
10^0
10^−1
(d)
(e)
(f)
η κ̂_t
2
1
0
0
1000
2000
3000
GD update t